MC 501 Definitions Practice: Worked Example

- Rebuilt from my own research project, Play to Learn in VR (P01.01), as the student study the Research Proposal example set up: one coder reading 150 tweets from 2021 about playing in VR.
- Everything under "The example" is what a student would submit. Edge cases are paraphrased from tweets in the archive; none is quoted.
- The rubric map at the end shows which grading row each part earns and why.

The example

Study	**Learning Talk and VR Tone: Affordances and Constraints in Tweets About Playing in Virtual Reality**
Unit of analysis	One English-language tweet, not a retweet, posted in 2021, that mentions VR together with play. Every variable below is coded on one tweet.

Variable 1: Learning reference

Part	**Learning reference**
Conceptual definition and source	Whether the tweet is about someone gaining or passing on a skill or knowledge in or through VR: learning, teaching, studying, practicing, or being trained. This definition is my own. The published discourse studies defined their learning tweets by a word list (Lampropoulos et al., 2022), and I define the variable by what the tweet describes instead, because in this archive the same words appear in game titles, adult-game advertisements, and idioms, and a word list cannot tell those apart from a person learning.
Operational definition	Read the whole tweet. Mark 1 if any clause describes a person (the writer, a named person, learners in general, or a customer in an advertisement) learning, teaching, practicing, studying, or being trained in something, and the activity happens in or through VR. Otherwise mark 0. If the tweet contains a learning word, check what the word means in its sentence before marking.
Level of measurement, and what it allows	Nominal: two named categories with no order. It allows counts and percentages and a chi-square test against another nominal or ordinal variable. It rules out a mean, because 0 and 1 are labels and not amounts.
Coded values	0 = no learning reference. 1 = learning reference.

Part	Learning reference
Edge-case decision rules	1. A learning word that is part of a game or product title (a shooting game called a trainer, an app called a trainer) or part of an idiom (hype train, learned the hard way, learned my lesson) is coded 0 unless another clause describes learning. 2. "Learn" meaning "find out about" (learn which headset games were played most this year) is coded 0. 3. A tweet that only shares a video title is coded 1 when the title describes someone teaching or learning in VR (a video in which one avatar teaches another how to do something in a social VR app), because the unit is the tweet text and the text describes learning. 4. An advertisement for a product that teaches (a VR app that teaches you to play the piano) is coded 1. The variable records that learning is described, not who describes it or why. 5. Machine-learning and artificial-intelligence hashtags are coded 0: the learning is a computer's, not a person's.
Validity sentence	I would doubt this variable if most of the tweets coded 1 turned out to be companies advertising learning apps, because it would then be measuring marketing copy rather than people describing learning.

Variable 2: Tone toward VR

Part	Tone toward VR
Conceptual definition and source	The writer's evaluation of VR or of the VR experience the tweet describes: whether the tweet speaks well of it, badly of it, or neither. This is sentiment in the sense the discourse studies use, the positive or negative evaluation carried by the words of a text (Lampropoulos et al., 2022; Mohammad & Turney, 2013), read here by a person rather than counted by a lexicon.
Operational definition	Find every word, phrase, or emoji in the tweet that evaluates VR, a VR headset, or a VR experience (fun, love, amazing; nauseous, broken, waste of money). Mark 3 when every evaluation is favorable, 1 when every evaluation is unfavorable, and 2 when the tweet has both or has none. Evaluations of something other than VR (a friend, a game that is not in VR, a public figure) do not count.
Level of measurement, and what it allows	Ordinal: three categories in order from negative to positive, with no claim that the steps are equal. It allows ordered comparisons (a larger share of learning tweets sits at positive), a median and a mode, and a chi-square test. It rules out a mean, because the distance from negative to neutral is not known to equal the distance from neutral to positive.
Coded values	1 = negative. 2 = neutral or mixed. 3 = positive.

Part	Tone toward VR
Edge-case decision rules	1. A tweet that praises a game played outside VR, a streamer, or a friend but says nothing evaluative about VR itself is coded 2. 2. An advertisement that lists features or a release date with no evaluative word is coded 2. An advertisement that calls the experience amazing is coded 3, because the variable records what the tweet says, not whether the writer is sincere. 3. A complaint the writer laughs off (a laughing emoji after describing a glitch that threw them out of a match) is coded by its literal words, so it is 1 unless a favorable clause also appears, which makes it 2. 4. Emoji count as evaluations: a heart or a fire emoji attached to VR is favorable, and a vomiting face is unfavorable.
Validity sentence	I would doubt this variable if the positive codes were mostly promotional tweets, because advertising language is favorable by design and would say nothing about how users feel about VR.

Variable 3: Affordance named

Part	Affordance named
Conceptual definition and source	The benefit the tweet attributes to VR: something VR lets a person do, or do better, in the writer's account. In affordance theory an affordance is what an environment offers a person to do (Gibson, 1979), and the categories below follow the benefits users and students name most often, engagement, interaction, collaboration, and skill practice (Allcoat & von Mühlenen, 2018; Laine & Lee, 2024).
Operational definition	Look for a clause that says what VR, a headset, or a VR experience lets someone do or makes better. Match that clause to one category below. If the tweet names benefits in two categories, mark the one that takes up more of the tweet's words; if they are equal, mark the one that appears first. If no clause names a benefit, mark 0.
Level of measurement, and what it allows	Nominal: named categories with no order. It allows counts, percentages, and a chi-square test. It rules out a mean and a median, because the categories cannot be ranked.
Coded values	0 = no benefit named. 1 = fun or engagement (enjoyment, excitement, being absorbed). 2 = hands-on interaction (moving, building, handling objects, being inside the scene). 3 = doing it with others (playing, meeting, or learning alongside other people). 4 = practicing a skill (getting better at something, in a game or outside it, in a safe setting). 5 = other benefit.

Part	Affordance named
Edge-case decision rules	1. A feature named without saying what it lets a person do (multiplayer, hand tracking, a high refresh rate) is coded 0. A feature is not a benefit until the tweet says what it is for. 2. "Fun" or "love it" with no more detail is coded 1. 3. Learning to play a VR game better is coded 4, the same as practicing piano in VR, because the variable records the kind of benefit, not whether it carries outside the game. 4. A tweet that praises VR for company is coded 3 only when other people are involved; a solo experience described as absorbing is coded 1.
Validity sentence	I would doubt this variable if more than a third of the tweets that name a benefit landed in "other benefit," because the categories would then not be describing what users actually name.

Variable 4: Constraint named

Part	Constraint named
Conceptual definition and source	The obstacle the tweet attributes to VR: something that stopped, limited, or discouraged the writer or someone else from doing something in VR. A constraint is this study's counterpart to an affordance (Gibson, 1979), what the environment blocks, and the categories follow the barriers the reviews report most often, cybersickness, accessibility, and hardware (Sunardi et al., 2025; van der Meer et al., 2023).
Operational definition	Look for a clause that names a problem with VR, the hardware, or the experience that limited or discouraged use. Match it to one category below. If the tweet names problems in two categories, mark the one that takes up more of the tweet's words; if they are equal, mark the one that appears first. If no clause names a problem, mark 0.
Level of measurement, and what it allows	Nominal: named categories with no order. It allows counts, percentages, and a chi-square test. It rules out a mean and a median.
Coded values	0 = no obstacle named. 1 = physical discomfort (motion sickness, nausea, eye strain, headache, fatigue). 2 = hardware cost (the price of a headset or computer, cannot afford one). 3 = technical performance (lag, crashes, glitches, setup trouble, incompatibility). 4 = other obstacle (space, time, other players' behavior, anything else).

Part	Constraint named
Edge-case decision rules	1. A problem with something other than VR (a slow phone, a friend's lost username) is coded 0 unless the tweet ties the problem to VR. 2. A warning or advice about a problem (how to build up tolerance to motion sickness) is coded in that category even when the writer did not suffer it, because the tweet names it as an obstacle. 3. Being thrown out of a match by a glitch is coded 3; what the writer lost as a result does not change the category. 4. "I can't play, no headset yet" is coded 2 only when price or affordability is stated; otherwise it is coded 4.
Validity sentence	I would doubt this variable if I only ever coded an obstacle when the tweet's tone was negative, because the variable would then be copying tone rather than recording which obstacle was named.

References

- Allcoat, D., & von Mühlenen, A. (2018). Learning in virtual reality: Effects on performance, emotion and engagement. *Research in Learning Technology, 26*. https://doi.org/10.25304/rlt.v26.2140
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Houghton Mifflin.
- Laine, T. H., & Lee, W. (2024). Collaborative virtual reality in higher education: Students' perceptions on presence, challenges, affordances, and potential. *IEEE Transactions on Learning Technologies, 17*, 280-293. https://doi.org/10.1109/TLT.2023.3319628
- Lampropoulos, G., Keramopoulos, E., Diamantaras, K., & Evangelidis, G. (2022). Augmented reality and virtual reality in education: Public perspectives, sentiments, attitudes, and discourses. *Education Sciences, 12*(11), Article 798. https://doi.org/10.3390/educsci12110798
- Mohammad, S. M., & Turney, P. D. (2013). Crowdsourcing a word-emotion association lexicon. *Computational Intelligence, 29*(3), 436-465. https://doi.org/10.1111/j.1467-8640.2012.00460.x
- Sunardi, Meyliana, Hendric, S. W. H. L., & Rosmansyah, Y. (2025). Exploring immersive virtual reality in higher education: Research gap and future direction, a scoping review. *IEEE Access, 13*, 76308-76321. https://doi.org/10.1109/ACCESS.2025.3565385
- van der Meer, N., van der Werf, V., Brinkman, W.-P., & Specht, M. (2023). Virtual reality and collaborative learning: A systematic literature review. *Frontiers in Virtual Reality, 4*, Article 1159905. https://doi.org/10.3389/frvir.2023.1159905

Rubric map

Points are the rubric's split of the 25: conceptual definition with its source 3, operational definition 6, level of measurement and what it allows 3, coded values 3, edge-case decision rules 5, validity sentence 5. Each row looks across all four variables.

Rubric row	Points	Where the example earns it	Why it earns the full row
Conceptual definition, with its source	3	The first row of each variable's table	Each one says what the variable captures in the abstract and cites where the idea comes from. The one invented idea, learning reference, gets a sentence defending why it departs from the published word-list definition. None is a list of examples.
Operational definition	6	The second row of each variable's table	Each one tells the coder what to do (read, find, match, mark) and how to resolve a tie, so a stranger could follow it and reach the same answer. None restates the conceptual definition in other words.
Level of measurement, and what it allows	3	The third row of each variable's table	Named for all four and matched to the values: the ordered set, tone, is the one ordinal variable. Each carries a sentence on what the level allows in analysis and what it rules out, including no mean of a nominal or ordinal variable.
Coded values	3	The fourth row of each variable's table	Every value is listed with what it means, and every tweet fits exactly one value, because the category variables carry a none value and an other value and the operational definition settles ties.
Edge-case decision rules	5	The numbered rules in the fifth row of each variable's table	Four or five rules per variable, above the minimum of two. Every rule is an instruction a coder can apply to the next tweet, not an intention, and each comes from a case that occurs in the archive: game titles with a learning word, advertisements, idioms, laughed-off complaints, features named without a purpose, and warnings about motion sickness.
Validity sentence	5	The last row of each variable's table	One sentence per variable, each naming a specific pattern in the coded data that would show the variable is measuring something other than its conceptual definition. No sentence is reused across variables.

Four variables done fully is the assignment. Every rule above is phrased so it could be pasted into the Extended Codebook unchanged.